Examiner only

5. A 30 kW gas boiler is used to provide heating and hot water in a house.

The energy released in one year by burning the gas is 22 000 kWh.

(a) Use the equation:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the mean time of use **per day** of the gas boiler. [4]
(1 year = 365 days)

time of use per day = ... h

(b) A 3 m^2 solar water heating system costs £3600 and it is estimated to reduce gas consumption to 20 500 kWh per year.

Gas costs 12 p per kWh.

Use the equation:

$$\text{cost} = \text{units used} \times \text{cost per unit}$$

to calculate the time to payback the cost of the solar water heating system. [4]

payback time = ... years

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